
Session 5 (2h) — 통신 (Serial) & Integration Capstone

Covers: 통신, 시리얼 통신 (on-site task) — plus ties together everything from Sessions 1–4 (아두이노 회로 구성 + 센서 연동, MCU/GPIO/ADC/PWM/인터럽트)

1. Serial (UART) communication — ~40 min

- `Serial.begin(9600)` opens a connection at 9600 baud (bits/sec) between the Arduino and your computer over USB.
- **Sending data out:** `Serial.print()` / `Serial.println()` — you’ve already used this for debugging in every session so far.
- **Receiving data in:** reading commands typed into the Serial Monitor lets your PC control the Arduino:

```
if (Serial.available() > 0) {  
  char command = Serial.read();  
  if (command == '1') { /* do something */ }  
}
```

- This is the actual 시리얼 통신 (serial communication) on-site task — sending sensor readings out, and/or receiving commands in, over the same USB link used for programming.

2. Why this matters for the internship — ~10 min

현장 업무 lists three things for this row: 아두이노 회로 구성 / 센서 연동 / 시리얼 통신. By the end of this session you will have touched all three in one working system — this is exactly the shape of task you’ll be asked to do on-site.

3. Capstone integration — ~70 min

`src/integrated_sensor_system.ino` combines every session’s content into one sketch: - **GPIO** — a status LED (Session 2) - **ADC** — reads a sensor/potentiometer on A0 (Session 3) - **PWM** — drives a second LED’s brightness from that sensor reading (Session 3) - **Interrupt** — a button on D2 toggles a “manual override” mode instantly (Session 4) - **Serial communication** — continuously reports sensor value + mode over Serial, and accepts a 'm' command typed into Serial Monitor as an alternate way to toggle override

This mirrors a realistic embedded task: read a sensor, react to it, respond instantly to a manual input, and report status back over serial — the same pattern used in real 아두이노 회로 구성 + 센서 연동 + 시리얼 통신 work.

Wiring for this session

Component	Pin	Other end
Status LED (+)	D8	220Ω resistor → anode
Status LED (–)	GND	cathode
PWM LED (+)	D9 (~PWM)	220Ω resistor → anode
PWM LED (–)	GND	cathode
Potentiometer wiper	A0	—
Potentiometer outer pins	5V and GND	one to each
Override switch leg A	D2	—
Override switch leg B	GND	—

See `schematic/circuit-diagram.svg` for the visual layout.

How to run it

1. Build in Tinkercad first (LED×2 + resistor×2 + potentiometer + pushbutton + Arduino Uno).
2. Paste in `src/integrated_sensor_system.ino`, start simulation.
3. Open Serial Monitor (9600 baud) — watch live sensor + mode reports.
4. Turn the potentiometer → PWM LED brightness follows it, values print over Serial.
5. Press the button (or type `m` + Enter in Serial Monitor) → override mode toggles instantly, confirmed both by the status LED and a Serial message.
6. Once confirmed in simulation, replicate on your physical breadboard.

Checkpoint — you're ready for the internship's HW prerequisite when:

- ☐ Can explain `Serial.begin/print/read` in your own words
- ☐ The capstone sketch runs correctly: sensor reading updates PWM LED, button/`m` command toggles override, everything reports over Serial
- ☐ Can point to where each of MCU/GPIO/ADC/PWM/interrupt/communication happens in the capstone code without help